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MatForge

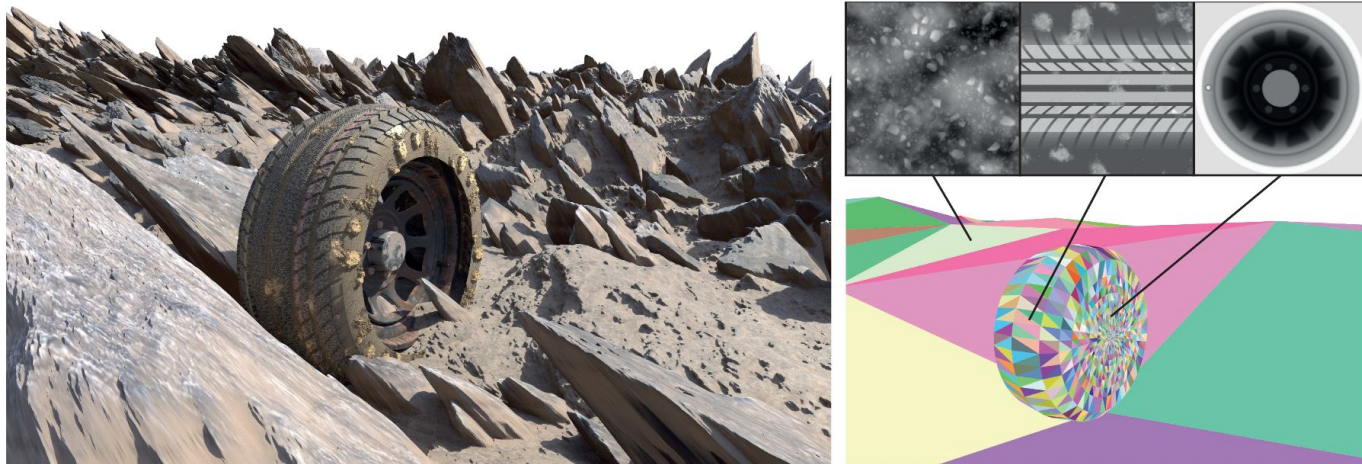
Milestone 3

Dec 1, 2025

Recap

MatForge is an advanced material rendering system. The project implements four complementary SIGGRAPH papers (2023-2024) in a unified Vulkan-based path tracing pipeline:

QOLDS, FastMSX, Bounded-VNDF, **RMIP**



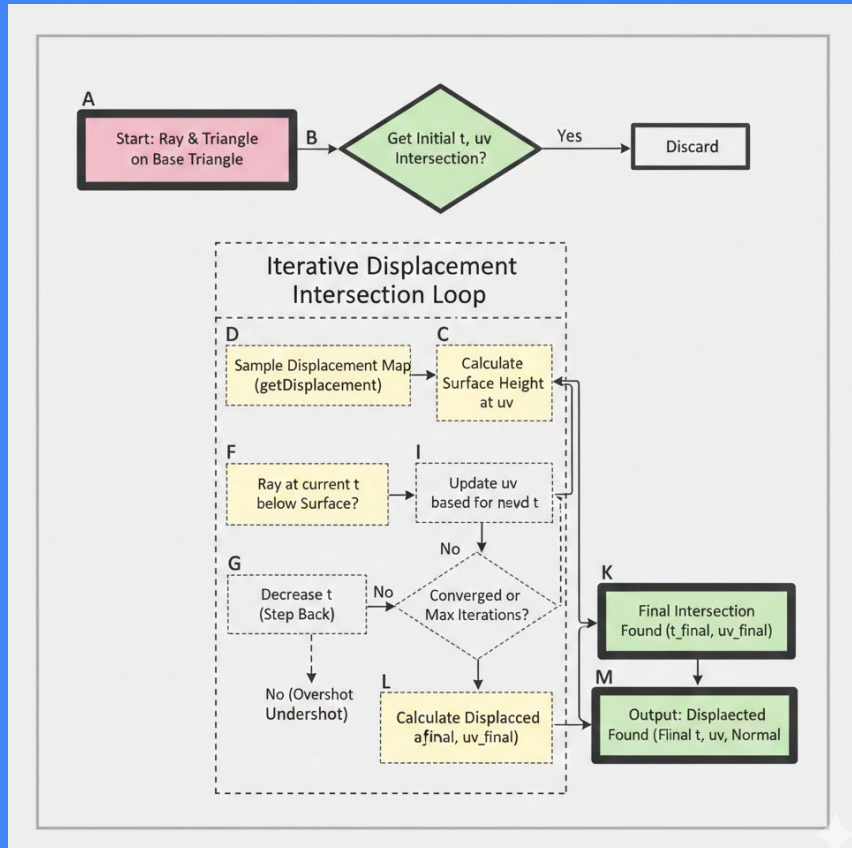
Milestone 3

RMIP

BLAS - AABB

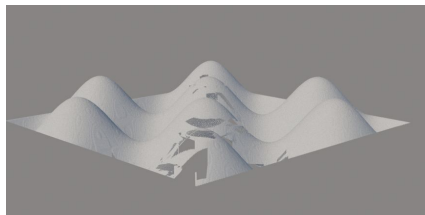
Displacement Intersection Shader

The paper's elegant ψ -guided marching is mathematically beautiful but numerically fragile in practice.

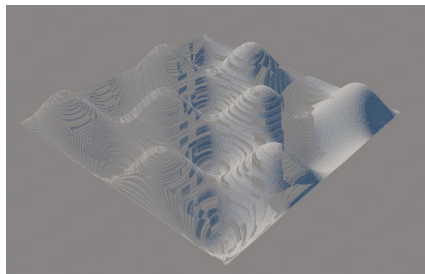
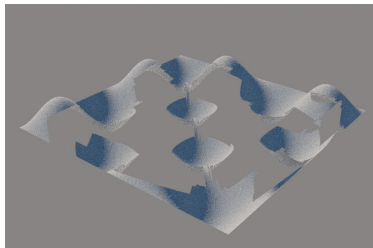


Summary of Fixes (V1 → V24)

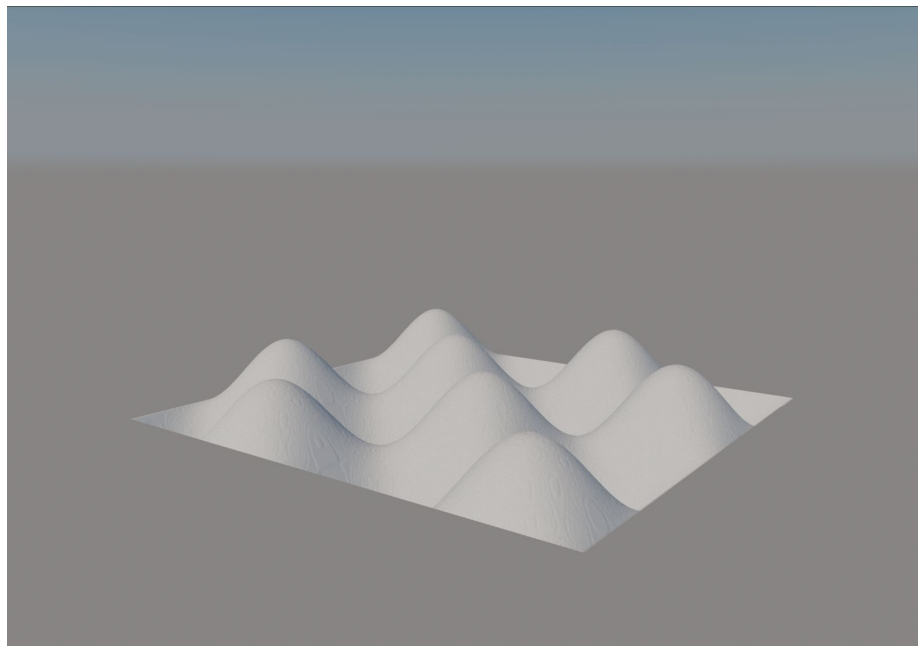
- ✓ Visually correct at all viewing angles
- ✓ Hierarchical traversal with RMIP queries working



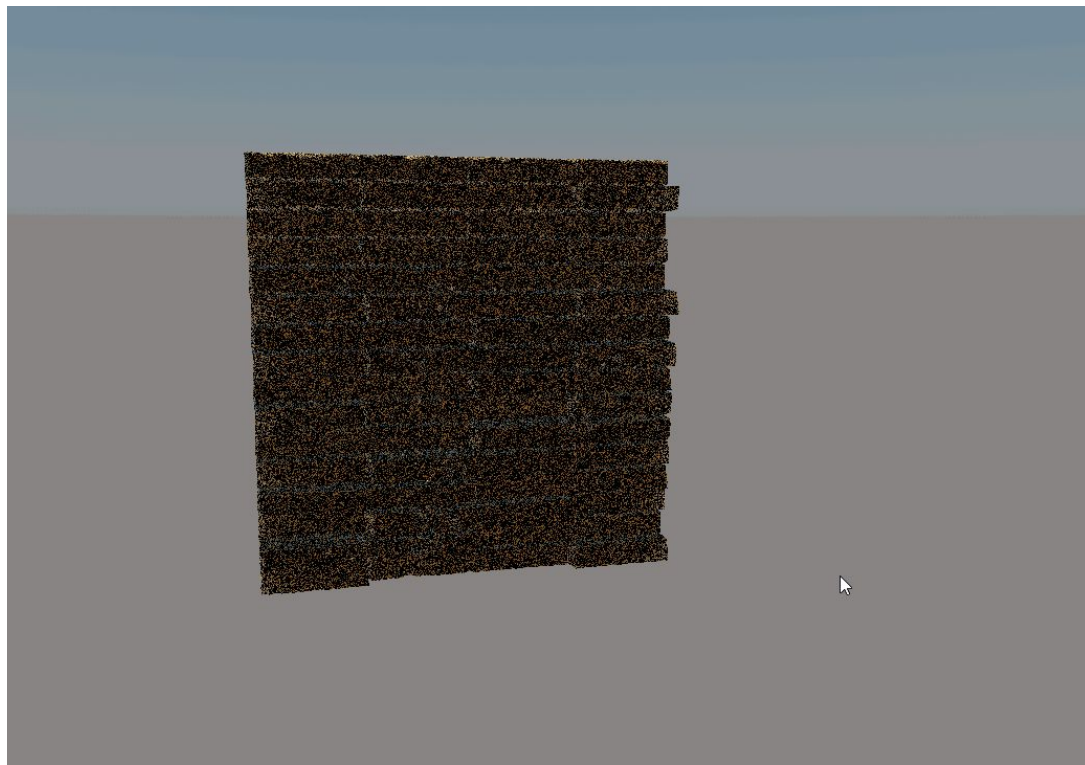
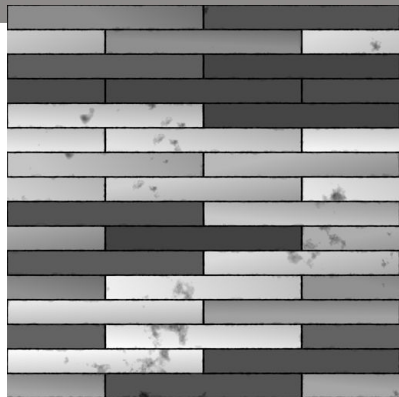
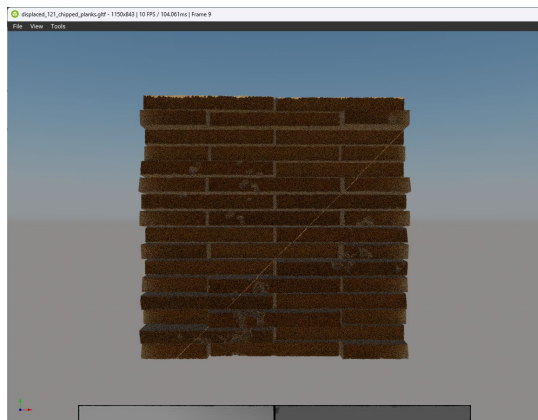
Invalid Inverse
Displacement
Results



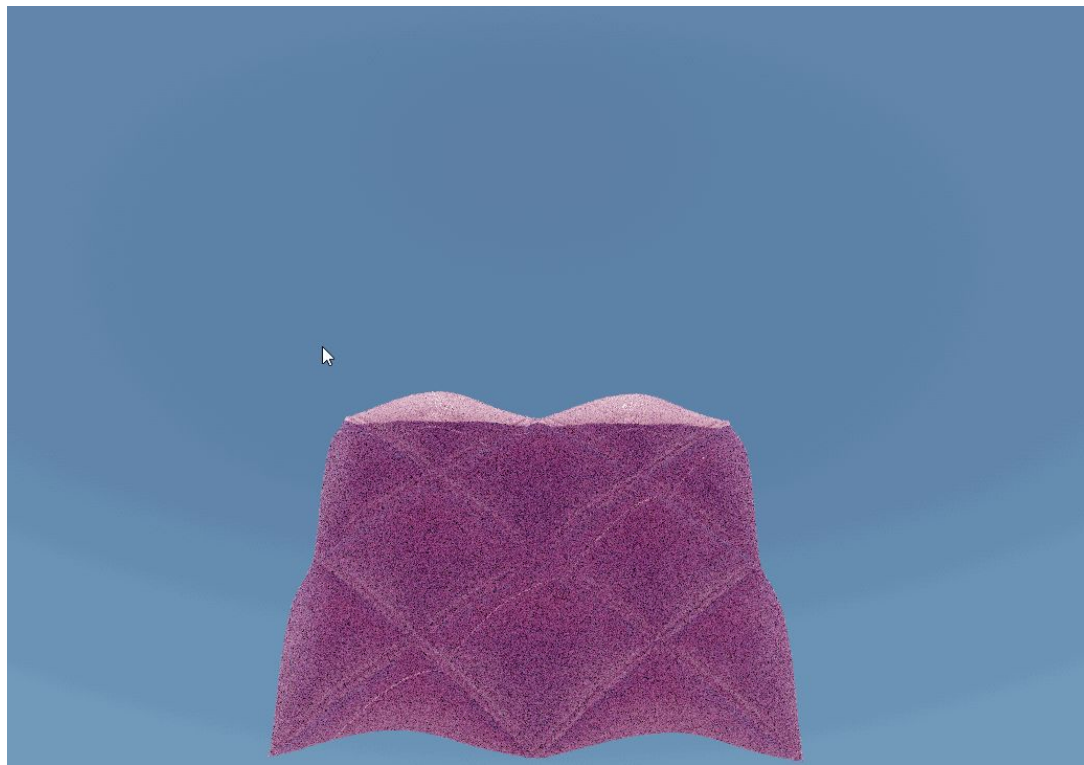
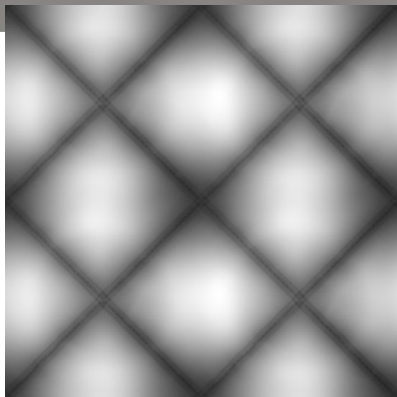
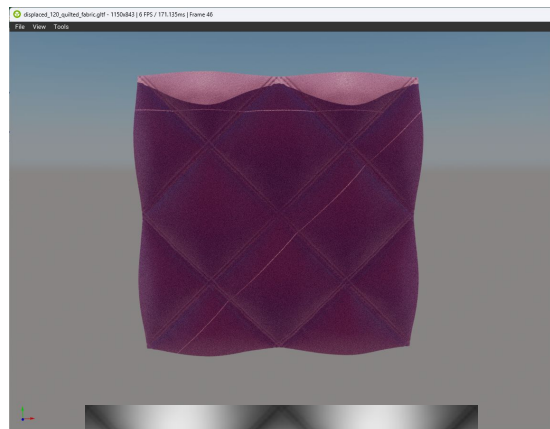
Severe Stripping
Artifacts



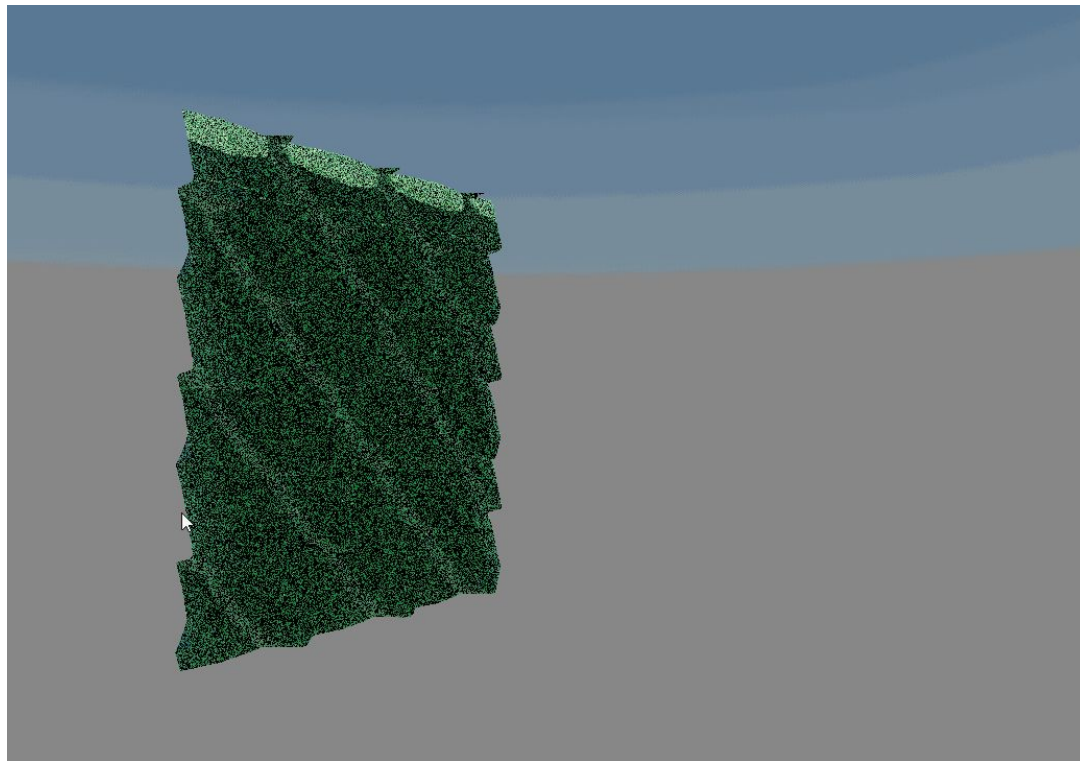
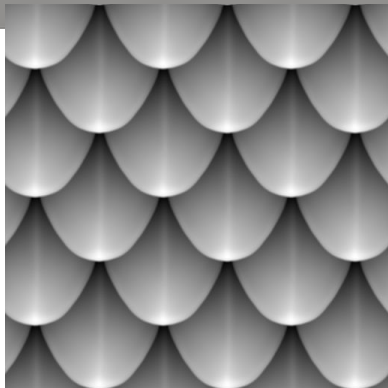
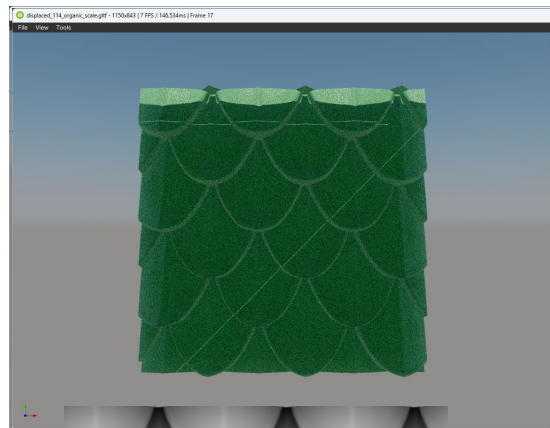
Test on RMIP



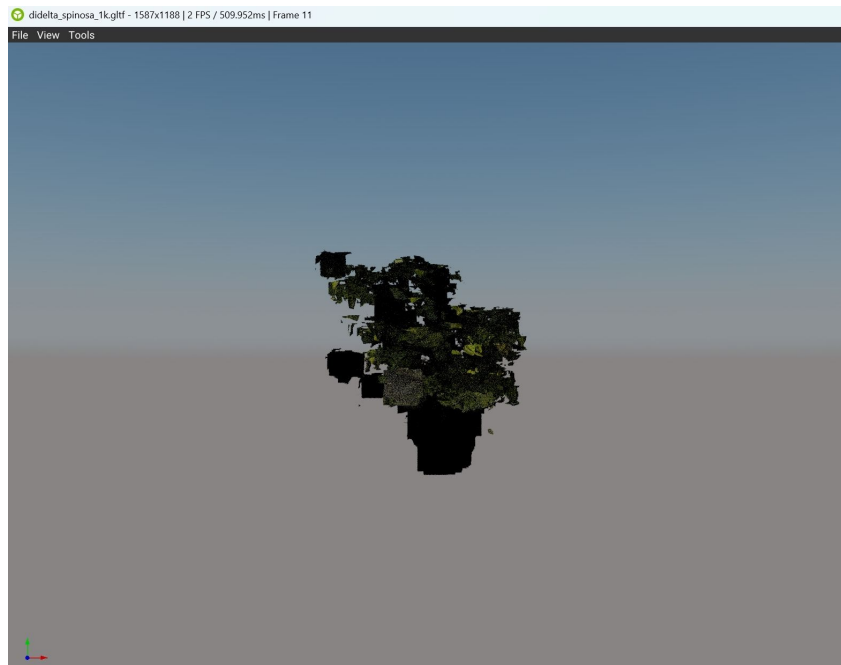
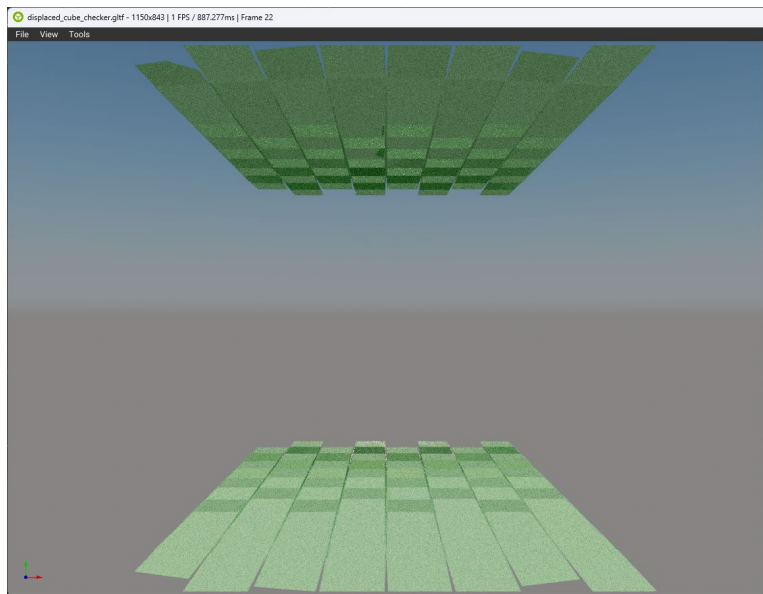
Test on RMIP



Test on RMIP



Displacement Mapping rendering errors in 3D objects



Milestone 3

The Demo Scene

Render the scene with
QOLDS, FastMSX, Bounded-VNDF





The Demo Scene

The Demo Scene



The Demo Scene - Issue



The Demo Scene - Issue



Camera

Scene Graph

Settings

Active Renderer

Debug Method

Path Tracer

None

Path Tracer

Rendering Technique

Ray Tracing

Use SDR

Max Depth

Firefly Clamp

Use QOLDS

Use FastMSX

Use Bounded VNDF

Max Iterations

Samples

Auto SPP

Performance Target

Balanced (30 FPS)

Samples: 434/434 (1.0x)

Throughput: 176.75 MSPPP/s (7)

Aperture

Auto Focus

Focal Distance

Infinite Plane

Enable DLSS

DLSS Size Mode

Current Resolution: 2075 x 1350

Guide Images

Environment

Tonemapper

Statistics

Convergence Analysis

Properties

Node: Plane.002

Translation

Rotation

Scale

Add Visibility

Material Selection:

Primitive 0 (patterned_paving_02)

Material: patterned_paving_02

Base Color

Metallic

Roughness

Emissive

Alpha Cutoff

Alpha Mode

Double Sided

Anisotropy

Clearcoat

Diffuse Transmission

Dispersion

Emissive Strength

IOR

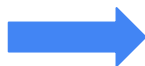
Irrescence

Before final presentation

- Resolve visual artifacts in 3D objects displacement mapping
- Optimize the performance of RMIP
- Create RMIP-specific performance analysis document with graphs
- Create performance graphs and visual comparisons

Polish & Documentation

- Show complete integrated system
- Complete material editor with live preview
- Present all benchmarks and analysis



Demo Documentation Presentation

Live Demo